

EC21-3: Machine Learning Techniques			
Teaching Scheme: Theory Session: Total 45 Hours		Credit: 03	Examination Scheme: Internal (TH): 25 Marks External (TH) : 50 Marks Total :75 Marks
Prerequisites: Data Processing, EDA, Statistics, Python			
Course Objectives: - To Understand the Concept of Machine Learning - To Gain knowledge on Supervised and unsupervised Learning techniques. - To evaluation of learning algorithms and model selection - To Explore Knowledge of Semi-Supervised and reinforcement learning - To Analyze case studies to understand successful implementations and challenges faced in practical scenarios.			
Course Outcomes: On completion of the course, learners should be able to			
CO#	Cognitive Domain	Course Outcomes	
CO1	Understand	Describe the workflow of a machine learning project, including data pre-processing, model training, evaluation, and deployment.	
CO2	Apply	Apply the various algorithms of supervised and learning	
CO3	Apply	Apply the various algorithms of unsupervised learning	
CO4	Apply	Apply the fundamental algorithms in semi-supervised and reinforcement learning.	
CO5	Apply	Apply real-world applications of supervised and unsupervised learning across diverse domains.	
Unit No.	Contents		No of Sessions
1	Introduction of ML 1.1 Overview of Machine Learning 1.2. Machine Learning Vs Statistical Learning 1.3. Type of Machine Learning 1.4 Machine Learning Workflow 1.4.1 Problem Definition and Data Collection 1.4.2 Data Preparation and Preprocessing 1.4.3 Model Selection and Training 1.4.4 Model Evaluation and Validation 1.5 Key Concepts and Terminology 1.5.1 Features, labels, datasets 1.5.2 Training set, test set, validation set 1.5.3 Overfitting vs. underfitting 1.5.4 Bias-variance tradeoff		8
		10	

*Mapping of Course Outcomes for Unit 1:CO1			
2	Supervised Learning 2.1. Introduction to Supervised Learning 2.2. Linear Regression 2.2.1 Simple Linear Regression 2.2.2 Multiple Linear Regression 2.3. Classification 2.3.1 Introduction to Classification 2.3.2 Logistic Regression 2.3.3 Decision Trees 2.3.4 k-Nearest Neighbors 2.3.5 Support Vector Machines 2.3.6 Naive Bayes Classifier 2.4. Evaluation Metrics: Accuracy, Precision, Recall, F1-Score 2.5. Model Evaluation and Validation 2.5.1 Train/Test Split 2.5.2 Cross-Validation 2.5.3 Overfitting and Underfitting 2.5.4 Confusion Matrix *Python Implementation of Supervised machine learning algorithm using Scikit-Learn	25	12
*Mapping of Course Outcomes for Unit 2: CO2			
3	Unsupervised learning 3.1 Introduction to Unsupervised Learning 3.1.1 Definition 3.1.2 Purpose 3.1.3 Unsupervised Learning Approaches 3.1.4 Applications and scenarios where unsupervised learning is used 3.2 Clustering Algorithms- 3.2.1 K-means Clustering 3.2.2 Hierarchical Clustering 3.2.3 Density-based Clustering (DBSCAN) 3.3 Dimensionality Reduction Techniques 3.4 Case studies	25	10
*Mapping of Course Outcomes for Unit 3:CO3			
4	Semi Supervised and Reinforcement: 4.1 Introduction to Semi-Supervised Learning 4.1.1 Importance of Semi-Supervised Learning 4.1.2 Applications in real-world scenarios 4.2 Techniques in Semi-Supervised Learning 4.2.1 Self-Training	20	8

	4.2.2 Co-Training 4.2.3 Graph-Based Methods 4.2.4 Semi-Supervised Support Vector Machines 4.2.5 Generative Models (e.g., Gaussian Mixture Models) 4.3 Introduction to Reinforcement Learning 4.3.1 Key concepts: Agent, Environment, State, Action, Reward 4.3.2 Applications of Reinforcement Learning 4.4 Markov Decision Processes (MDPs) 4.5 Temporal-Difference Learning- SARSA and Q-Learning 4.6 Advanced Topics in Reinforcement Learning 4.6.1 Deep Q-Networks (DQN) (brief introduction) 4.6.2 Exploration vs. Exploitation Trade-off 4.6.3 Reinforcement Learning in complex environments		
*Mapping of Course Outcomes for Unit 4: CO4			
5	Case studies 5.1 REGRESSION Case Studies 5.1.1 Retail Store Sales Prediction 5.1.2 Restaurant Sales Prediction 5.1.3 Inventory Prediction for Optimum Inventory Management 5.1.4 Sport Player Salary Prediction 5.1.5 Machine Learning case study on Dell 5.2 CLASSIFICATION Case Studies 5.2.1 Diabetes Prediction for Preventive Care 5.2.2 Telecom Network Disruptions Prediction for Planning Preventive Maintenance 5.2.3 Breast Cancer Prediction for Preventive Care 5.2.4 Credit Card Fraud Detection 5.2.5 Heart Diseases Prediction for Preventive Care 5.2.6 Loan Application Classification 5.2.7 Computer Price estimation 5.2.8 House price prediction	20	7
*Mapping of Course Outcomes for Unit 5:CO5			
Learning Resources			
Text Books: • "Foundations of Machine Learning" by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar • Machine Learning: An Algorithmic Perspective" by Stephen Marsland • Statistics and Data Science -Paperback, Dr. Swapnaja, Dr. Minakshi, Dr. Mukul			

Kulkarni, Dr. Santosh Deshpande, Dr. Ravikant Zirmite

Reference Books:

- "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems" by Aurélien Géron
- "Pattern Recognition and Machine Learning" by Christopher M. Bishop
- "Reinforcement Learning: An Introduction" by Richard S. Sutton and Andrew G. Barto
- "Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython" by Wes McKinney

Recommended Learning Material:

1. Coursera: Machine Learning by Andrew Ng - <https://www.coursera.org/learn/machine-learning>
2. GitHub: Awesome Machine Learning - <https://github.com/josephmisiti/awesome-machine-learning>